

Please Grab a handout in the back.

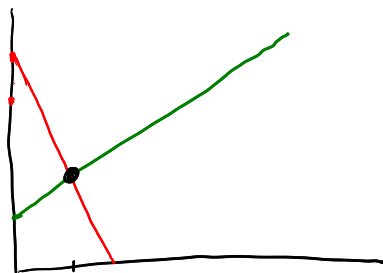
Today

Solving Nonlinear Function
Bracketing Methods: Most important: Bisection

Announcements / Reminders

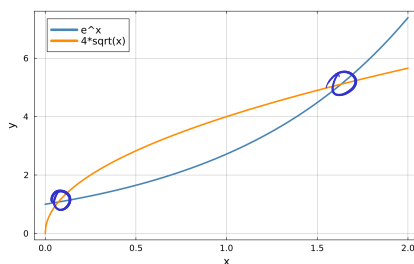
Lab 1 due Thursday - Check Auto-grader!
HW 1 Released (Graded)

Linear



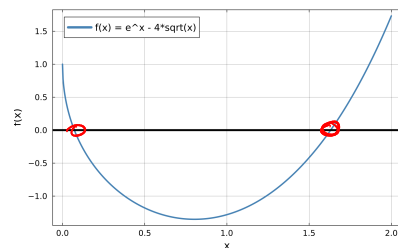
$$\begin{aligned} -2x + 4 &= x + 1 \\ x_2 &= -2x_1 + 4 & x_2 &= x_1 + 1 \\ x &= 1 \end{aligned}$$

Nonlinear



$$\begin{aligned} e^x &= 4\sqrt{x} \\ \text{no analytical in} \\ \text{"simple functions"} \end{aligned}$$

⇒



$$\begin{aligned} f(x) &= 0 = e^x - 4\sqrt{x} \\ \text{find "roots"} \\ x\text{-values where } f(x) &= 0 \end{aligned}$$

Bracketing

Open

↑ Today

Incremental Search

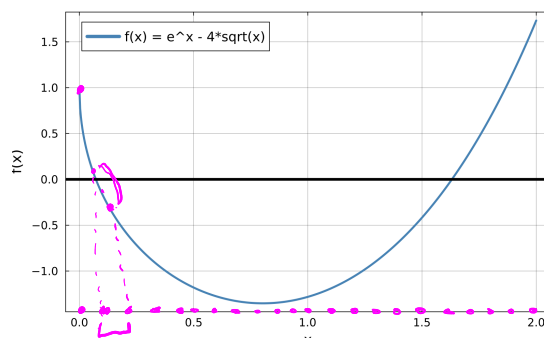
Incremental Search

input : function f , search range $[x_{\min}, x_{\max}]$, number of steps n
output: list of brackets $[x_l, x_u]$, each containing a root

```

 $\Delta x \leftarrow (x_{\max} - x_{\min}) / n$ 
brackets  $\leftarrow \{\}$ 
 $x_l \leftarrow x_{\min}$ 
for  $k \leftarrow 1$  to  $n$  do
   $x_u \leftarrow x_l + \Delta x$ 
  if  $f(x_l) \cdot f(x_u) < 0$  then
    // sign change ⇒ a root lies in  $[x_l, x_u]$ 
    append  $[x_l, x_u]$  to brackets
  end
   $x_l \leftarrow x_u$ 
end
return brackets

```



↑ if $f(x)$ is continuous and $f(x_l)$ and $f(x_r)$ have different signs, a root is between x_l and x_r

Strength: Can find multiple roots

Weakness: Can miss roots, slow at finding one root (we will see shortly)

error $\rightarrow \epsilon_a = \frac{\Delta x}{n-1}$

Bisection

Bisection

input : function f , bracket $[x_l, x_u]$ with $f(x_l) \cdot f(x_u) < 0$, tolerance ϵ_s
output: estimate x_r of the root

```

 $x_r \leftarrow x_l$ 
repeat
   $x_{r,old} \leftarrow x_r$ 
   $x_r \leftarrow (x_l + x_u) / 2$ 
  // keep the half that still brackets the root
  if  $f(x_l) \cdot f(x_r) < 0$  then
     $x_u \leftarrow x_r$ 
  else if  $f(x_l) \cdot f(x_r) > 0$  then
     $x_l \leftarrow x_r$ 
  else
    return  $x_r$  // exact root
  end
until  $\epsilon_a < \epsilon_s$ 
return  $x_r$ 

```

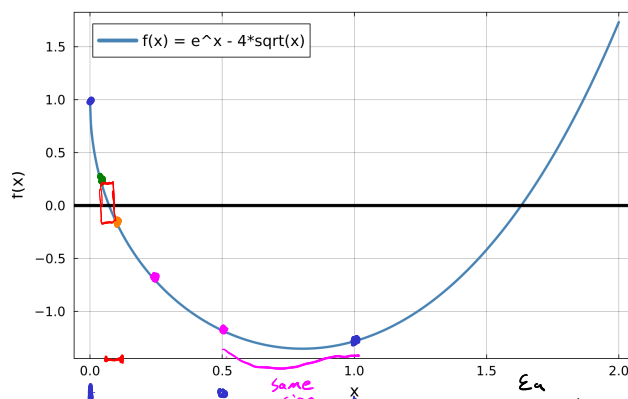
Iter: 1

2

3

4

5



0.0675 0.125

ϵ_a

$\Delta x = 1$

$\Delta x = 0.5$

$\Delta x = 0.25$

$\Delta x = 0.125$

$\Delta x = 0.0675$

$\epsilon_a = \frac{\Delta x}{2^n}$

$$\epsilon_a = \frac{\Delta x}{2^n}$$

function calls
for desired $\frac{\Delta x}{\epsilon_a}$

Incremental

$$\epsilon_a = \frac{\Delta x}{n-1}$$

$$n-1 = \frac{\Delta x}{\epsilon_a}$$

$$n = \frac{\Delta x}{\epsilon_a} + 1$$

$$O\left(\frac{\Delta x}{\epsilon_a}\right)$$

Bisection

$$\epsilon_a = \frac{\Delta x}{2^n}$$

$$2^n = \frac{\Delta x}{\epsilon_a}$$

$$n = \log_2\left(\frac{\Delta x}{\epsilon_a}\right)$$

$$O\left(\log\left(\frac{\Delta x}{\epsilon_a}\right)\right)$$

$$\epsilon_a = 0.01 \Delta x$$

$$\frac{\Delta x}{\epsilon_a} = 100$$

101 func evals

$$\log(100) = 6.64$$

7 function evals !

$$\epsilon_a = 0.0001 \Delta x$$

$$\frac{\Delta x}{\epsilon_a} = 10,000$$

10,001 func evals

14 func evals

False Position

False Position

input : function f , bracket $[x_l, x_u]$ with $f(x_l) \cdot f(x_u) < 0$,
tolerance ϵ_a

output: estimate x_r of the root

$x_r \leftarrow x_l$

repeat

$x_{r,old} \leftarrow x_r$

$$x_r \leftarrow x_u - \frac{f(x_u)(x_l - x_u)}{f(x_u) - f(x_l)}$$

// keep the side that still brackets the root

if $f(x_l) \cdot f(x_r) < 0$ then

$x_u \leftarrow x_r$

else if $f(x_l) \cdot f(x_r) > 0$ then

$x_l \leftarrow x_r$

else

 return x_r

end

$$\epsilon_a \leftarrow \left| \frac{x_r - x_{r,old}}{x_r} \right| \cdot 100\%$$

until $\epsilon_a < \epsilon_a$

return x_r

← only difference
from bisection

Problem: Doesn't
handle
curvature well

